

Nomor 1

Write your answers for each question on the answer sheets that are provided.

1. [LO3] As part of the machine learning process, it is necessary to perform data pre-processing, which involves missing values handling and features encoding. Assuming you are provided with the following data:
 - a. (10%) Explain the missing value handling method that you use and provide an overview of the data after the missing values have been handled!
 - b. (10%) Explain the feature encoding method for categorical data, and provide an overview of the data after being encoded.

Customer Name	Gender	Age	Payment Method	LastTransaction	Churn
Jeffrey Pierce	male	46	credit card	86	loyal
Marion Wade	female	71	credit card	27	loyal
Tim Harrison	male	68	cash	179	loyal
Grace Newton			cash	173	churn

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Nora Holmes	female	49	cash	188	churn
Tracy Osborne	female	23		118	loyal
Ricardo Jordan	male	38	credit card	131	loyal
Tammy Kennedy		65	cash	187	churn
Ellis Lloyd	female		credit card	72	loyal
Marvin Silva	male	21	credit card	89	loyal
Christine Simon	female	37		162	churn
Franklin Griffith	female	70	credit card	153	churn
Juan Barnett	male	81	cash	153	churn
Velma Montgomery			credit card	210	churn
Mamie Cox	male	55	cash	126	churn
June Patrick	male	42	credit card	83	loyal
Edgar Frank	male	38	credit card	114	loyal
Frances Benson		60		142	loyal
Billie Thompson	male		credit card	112	loyal
Roman Patterson	male	22	credit card	88	loyal

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2. [LO1, LO4] You have sample data regarding the patient's health condition, to predict whether the patient is healthy or Not healthy. The description of the sample data is as follows:

Patient	Fever	Vomiting	Shivering	Class
P1	no	no	no	Healthy (H)
P2	average	no	yes	Not Healthy (NH)
P3	high	no	yes	Not Healthy (NH)
P4	high	yes	no	Not Healthy (NH)
P5	average	no	no	Healthy (H)
P6	no	yes	no	Healthy (H)
P7	average	yes	no	Not Healthy (NH)

You need to perform machine learning modeling using a Decision Tree algorithm under the given conditions:

- Metrics: Cross entropy : $entropy = -\sum_{i=1}^m p_i \log p_i$, where m is value of a variable
 - Split mechanism: expanded until all leaves are pure or until all leaves contain less than 1 samples.
- a. [20%] Provide a detailed explanation of the steps you performed to construct a complete decision tree.
 - b. [5%] If the given data includes the attributes: shivering = 'no', temperature = 'high', and vomiting = 'yes', according to the decision tree that has been constructed, the resulting class associated with this data is...

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3. [LO2,LO3] The regression data in the table below consists of 7 data samples that are utilized as training data for the K-Nearest Neighbor (KNN) algorithm.
- [10%] Apply the MinMax Scaler technique to normalize the Age and Salary features.
 - [10%] Calculate the salary value for the 8th data point using the value K=3 and the Euclidean distance

$$d(w, v) = \sqrt{\sum_{i=1}^n (w_i - v_i)^2}$$

ID	Years Experience (year)	Age (year)	Salary (IDR)
1	1.2	30	5,901,600
2	1.4	22	6,930,900
3	1.6	32	5,659,800
4	2.1	25	6,528,900
5	2.3	33	5,983,800
6	3	40	8,496,450
7	3.1	37	9,022,650
8	3.3	29	?

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4. [LO4] [20%] The outcomes of the segmentation classification model are displayed in the predicted segmentation column in the table provided below. Determine the values of Accuracy, Precision, Recall, and F1 Score. Please Provide the confusion matrix .

ID	Gender	Ever_Married	Graduated	Profession	Spending_Score	Family_Size	Actual Segmentation	Predicted Segmentation
462809	Male	No	No	Healthcare	Low	4	B	B
462643	Female	Yes	Yes	Engineer	Average	3	C	A
466315	Female	Yes	Yes	Engineer	Low	1	B	B
461735	Male	Yes	Yes	Lawyer	High	2	C	C
462669	Female	Yes	Yes	Entertainment	High	6	A	A
461319	Male	Yes	No	Artist	Average	2	D	C
460156	Male	No	Yes	Healthcare	Low	3	A	B
464347	Female	No	Yes	Healthcare	Low	3	D	D
465015	Female	Yes	Yes	Engineer	Low	3	B	B
465176	Female	Yes	Yes	Artist	Average	4	A	A
464041	Female	No	Yes	Engineer	Low	3	A	B
464942	Male	No	No	Healthcare	Low	4	C	C
461230	Female	No	No	Executive	Low	1	B	B
459573	Male	Yes	No	Lawyer	Low	1	A	A
460849	Female	Yes	No	Doctor	Low	1	D	D
460563	Female	No	No	Healthcare	Low	2	A	A
466865	Female	No	Other	Homemaker	Low	5	B	C
461644	Male	No	No	Healthcare	Low	6	C	C
466772	Male	Yes	Other	Entertainment	Average	4	B	B
464291	Female	Yes	Other	Artist	High	1	A	C
466084	Male	Yes	Other	Homemaker	Low	1	A	A

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459675	Female	No	No	Healthcare	Low	4	D	D
465602	Male	Yes	Other	Artist	Low	2	C	C
459168	Female	No	Other	Artist	Low	2	B	B
461021	Female	Yes	No	Executive	Average	3	B	B
465083	Male	Yes	No	Artist	Average	3	A	A
467604	Male	No	No	Healthcare	Low	8	D	D
459717	Male	Yes	Yes	Artist	Average	3	A	A
465882	Male	No	Yes	Artist	Low	1	B	B
461996	Female	No	No	Artist	Low	4	C	C

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5. [LO1] [15%] You assist researchers in unexplored regions of Kalimantan's woods to categorize plant species based on many features, including their habitat, leaf color, root shape, stem shape, and fruit presence. If data were accessible for these features and you employed machine learning to conduct the categorization, which methodology would you select and why? If you determine that dimension reduction is necessary based on your findings, what methodology will you employ?